

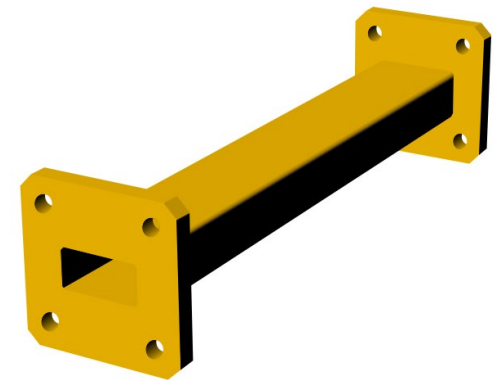
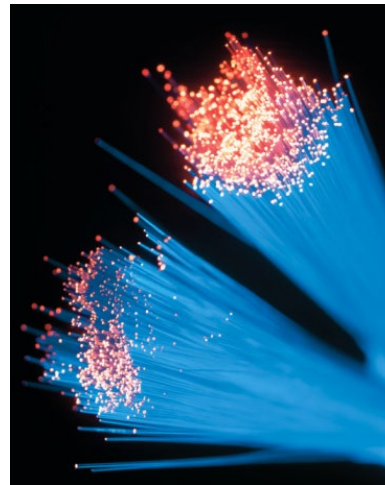
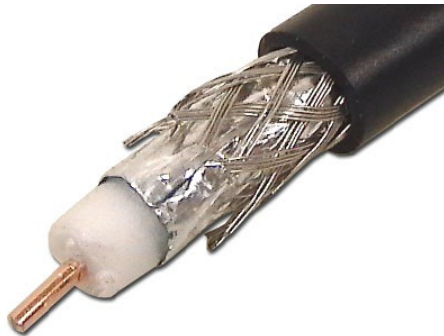
ECE 3317

Applied Electromagnetic Waves

Prof. David R. Jackson
Fall 2025

Notes 19

Waveguiding Structures



Waveguiding Structures

A waveguiding structure is one that carries a signal (or power) from one point to another without having energy escape.

There are three common types:

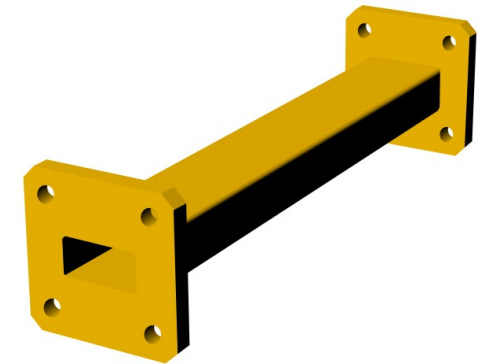
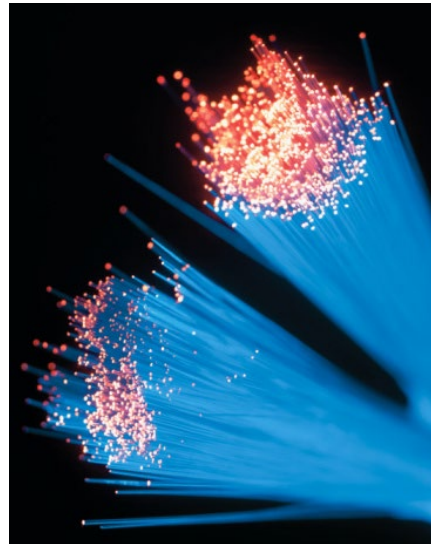
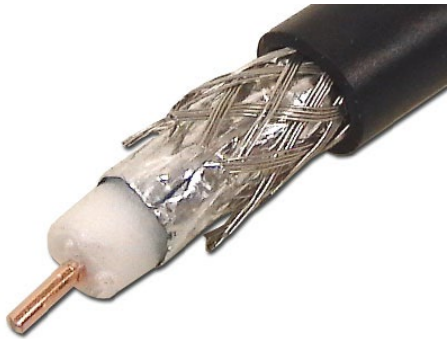
- Transmission lines
- Fiber-optic guides
- Rectangular waveguides (hollow pipes)

An alternative to a waveguiding system is a wireless system using antennas.

Waveguiding Structures (cont.)

Three common types of waveguiding structures:

- Transmission lines
- Fiber-optic guides
- Rectangular waveguides (hollow pipes)



Transmission lines

Properties:

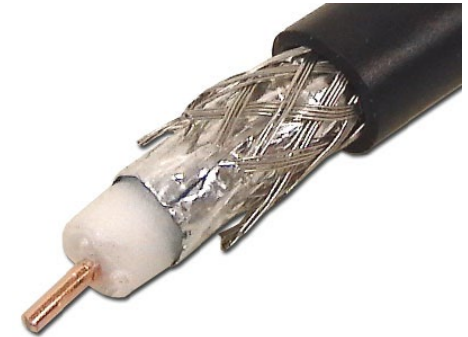
- Has two conductors running parallel.
- Can propagate a signal at any frequency (in theory).
- Becomes lossy at high frequency.
- Can handle low or moderate amounts of power.
- Has signal distortion due to loss.
- May or may not be immune to interference.
- Does not have E_z or H_z components of the fields (TEM_z).

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$k_z = -j\gamma = \beta - j\alpha$$

Lossless:

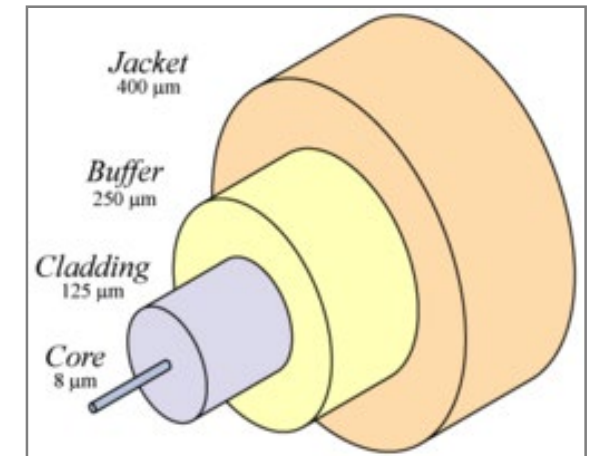
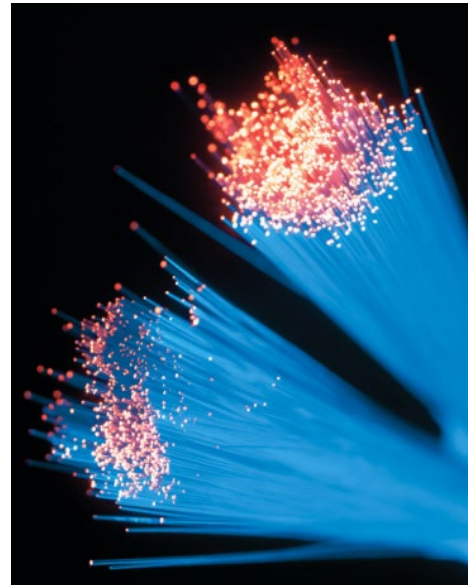
$$k_z = \omega\sqrt{LC} = \omega\sqrt{\mu\epsilon} = k = k_0\sqrt{\epsilon_r} \quad (\text{nonmagnetic filling})$$



Fiber-Optic Guide

Properties:

- Has a single dielectric rod.
- Can propagate a signal at any frequency (in theory).
- Can be made very low loss (no metal + low-loss glass).
- Has minimal signal distortion.
- Very immune to interference.
- Not suitable for high power.
- Has both E_z and H_z components of the fields (“hybrid mode”).



Fiber-Optic Guide (cont.)

Two types of fiber-optic guides:

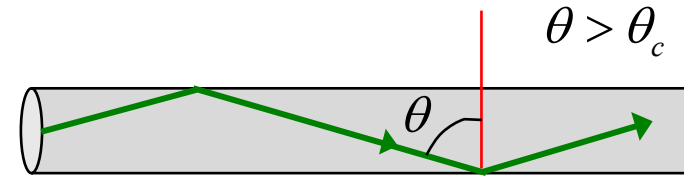
1) Single-mode fiber

Carries a single mode. Requires the fiber diameter to be small relative to a wavelength. This is the lowest loss type of fiber.

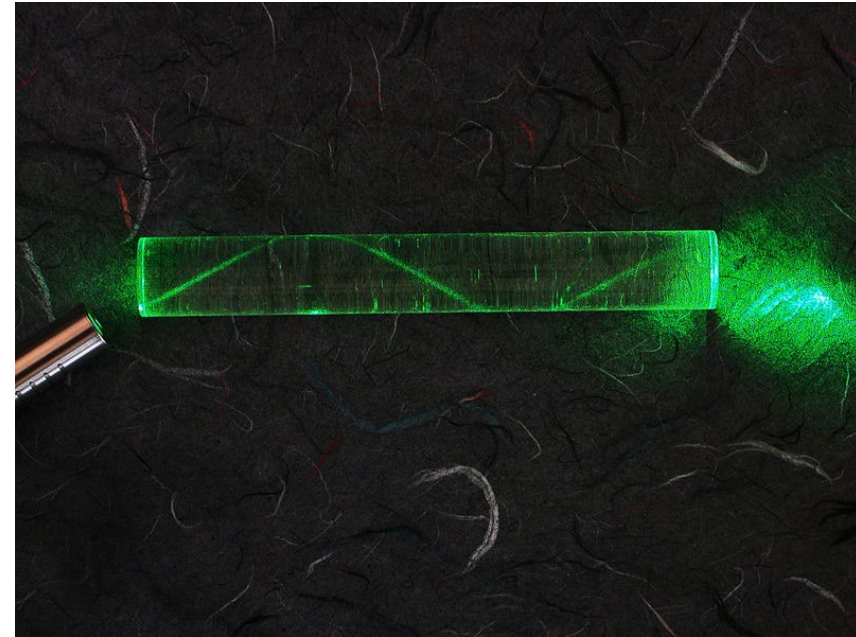
2) Multi-mode fiber

Has a fiber diameter that is large relative to a wavelength. It operates on the principle of total internal reflection (critical angle effect).

Multi-Mode Fiber



The multi-mode fiber has a fiber diameter that is large relative to a wavelength. It operates on the principle of total internal reflection (critical angle effect).

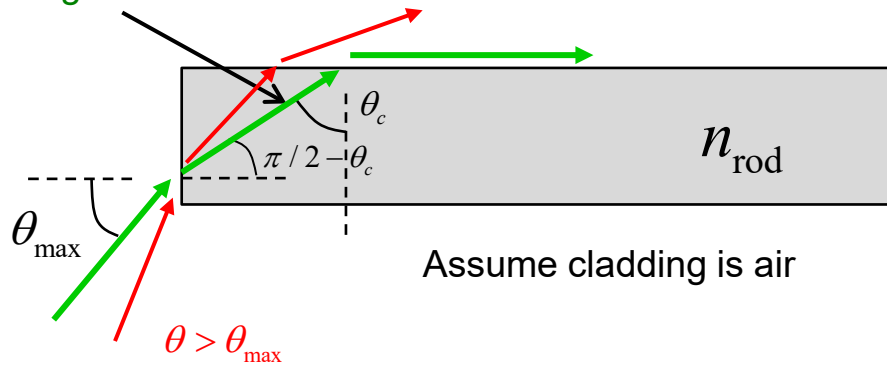


http://en.wikipedia.org/wiki/Optical_fiber

Multi-Mode Fiber (cont.)

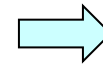
There is a maximum angle $\theta_{\max}$ at which the beam can be incident from outside the rod.

At the critical angle

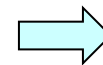


At left end of rod (input), Snell's law gives us:

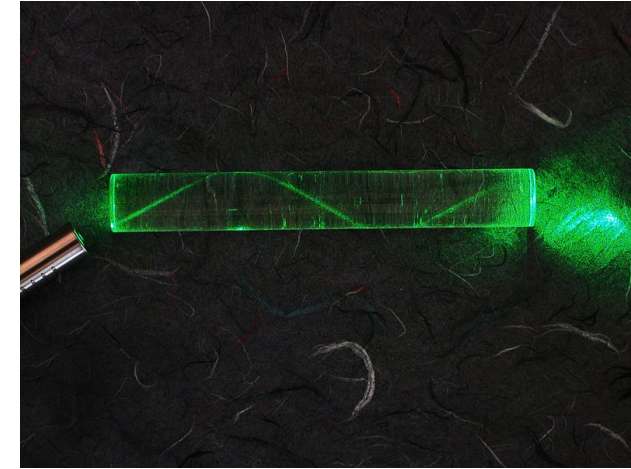
$$n_{\text{air}} \sin \theta_{\max} = n_{\text{rod}} \sin \left(\frac{\pi}{2} - \theta_c \right)$$



$$\sin \theta_{\max} = n_{\text{rod}} \cos \theta_c$$



$$\sin \theta_{\max} = n_{\text{rod}} \sqrt{1 - (1/n_{\text{rod}}^2)}$$



$$n_{\text{rod}} \sin \theta_c = n_{\text{air}} \sin(90^\circ) = (1) \sin(90^\circ) = 1$$

$$\Rightarrow \sin \theta_c = 1/n_{\text{rod}}$$

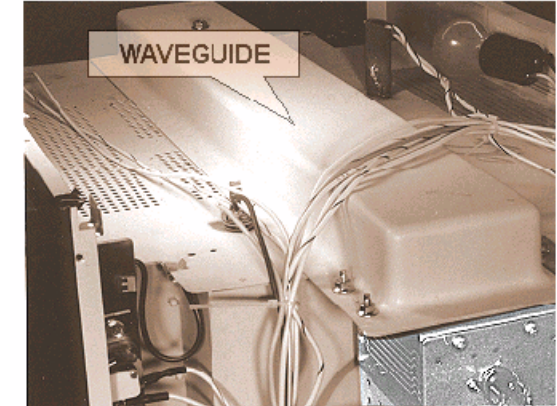
$$\Rightarrow \cos \theta_c = \sqrt{1 - \sin^2 \theta_c} = \sqrt{1 - (1/n_{\text{rod}}^2)}$$



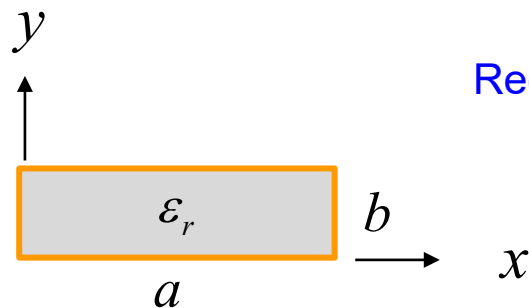
Rectangular Waveguide

Properties:

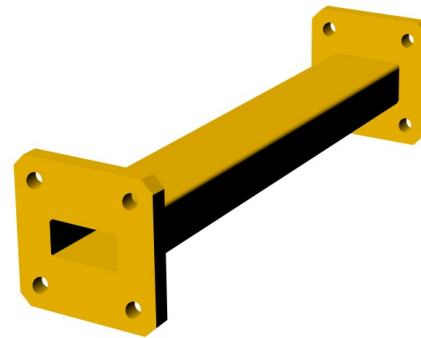
- Consists of a hollow rectangular metal pipe*.
- Can propagate a signal only at high frequency: $f > f_c$.
- Has signal distortion, even in the lossless case.
- Immune to interference.
- Can handle large amounts of power.
- Has low loss (compared with a transmission line).
- Has either E_z or H_z component of the fields (TM_z or TE_z).
- It usually operates in the TE₁₀ mode.



Inside microwave oven



Rectangular waveguide



***Note:** The pipe may be filled with a material having ϵ_r . ($\lambda_d = \lambda_0 / \sqrt{\epsilon_r}$)

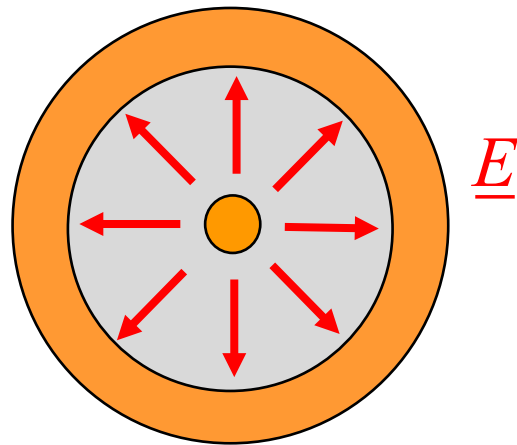
[https://en.wikipedia.org/wiki/Waveguide_\(radio_frequency\)](https://en.wikipedia.org/wiki/Waveguide_(radio_frequency))



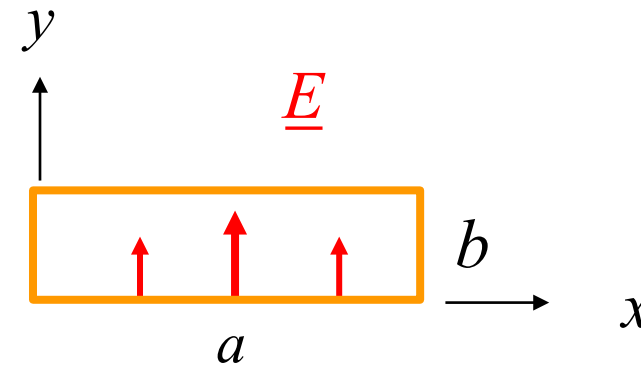
Rectangular Waveguide (cont.)

Rectangular Waveguide vs. Coax

- ❖ The coax has higher loss and is more susceptible to dielectric breakdown (can handle less power).
 - The electric field and surface current of the coax are concentrated near the narrow inner conductor (very strong electric field and current density there). This gives more loss and more susceptibility to dielectric breakdown.
 - The rectangular waveguide is usually hollow and thus has no dielectric loss.



Coax



Rectangular Waveguide

Waveguide (cont.)

Wavenumber inside a rectangular waveguide (derived in Appendix A):

$$k_z = \left(k^2 - k_c^2\right)^{1/2}$$

Note: k_c is different for each mode.

where

$$k = \omega\sqrt{\mu\epsilon} \quad (\text{wavenumber of material inside waveguide})$$

$$k_c = \pi / a \quad \text{for the dominant TE}_{10} \text{ mode}$$

Cutoff frequency f_c for a lossless waveguide (k is real):

f_c is the frequency for which $k = k_c$

$$f > f_c : \quad k_z = \sqrt{k^2 - k_c^2} = \text{real} \quad (\text{propagation})$$

$$f < f_c : \quad k_z = -j\sqrt{k_c^2 - k^2} = \text{imaginary} \quad (\text{evanescent decay})$$

Field Expressions for a Guided Wave

All six field components for a guided wave can be expressed in terms of the two fundamental field components E_z and H_z .

Assumption:

$$\begin{aligned}\underline{E}(x, y, z) &= \underline{E}_0(x, y) e^{-jk_z z} \\ \underline{H}(x, y, z) &= \underline{H}_0(x, y) e^{-jk_z z}\end{aligned}$$

(This is the definition of a guided wave.)

A derivation is given in Appendix B.

See the table on the next slide for the results.

Field Expressions for a Guided Wave (cont.)

Table of Fields

$$E_x = \left(\frac{-j\omega\mu}{k^2 - k_z^2} \right) \frac{\partial H_z}{\partial y} - \left(\frac{jk_z}{k^2 - k_z^2} \right) \frac{\partial E_z}{\partial x}$$

$$E_y = \left(\frac{j\omega\mu}{k^2 - k_z^2} \right) \frac{\partial H_z}{\partial x} - \left(\frac{jk_z}{k^2 - k_z^2} \right) \frac{\partial E_z}{\partial y}$$

$$H_x = \left(\frac{j\omega\varepsilon}{k^2 - k_z^2} \right) \frac{\partial E_z}{\partial y} - \left(\frac{jk_z}{k^2 - k_z^2} \right) \frac{\partial H_z}{\partial x}$$

$$H_y = \left(\frac{-j\omega\varepsilon}{k^2 - k_z^2} \right) \frac{\partial E_z}{\partial x} - \left(\frac{jk_z}{k^2 - k_z^2} \right) \frac{\partial H_z}{\partial y}$$

This table shows how to get all fields from E_z and H_z for any guided wave.

TEM_z Wave

Assume a TEM_z wave:

$$E_z = 0$$

$$H_z = 0$$

To avoid having a completely zero field (see table on previous slide):

$$k^2 - k_z^2 = 0$$

Hence,

$$\text{TEM}_z \iff k_z = k$$

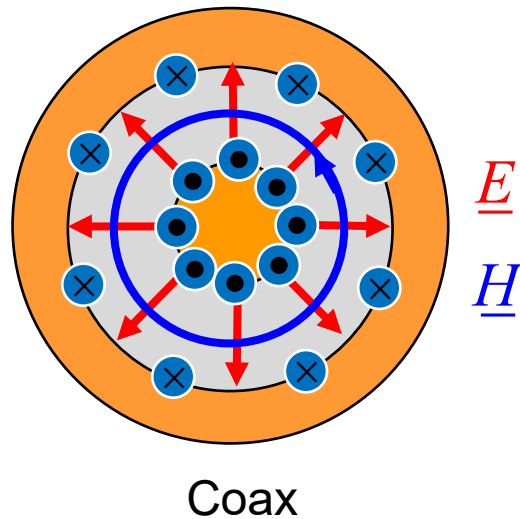
TEM_z Wave (cont.)

Examples of TEM_z waves:

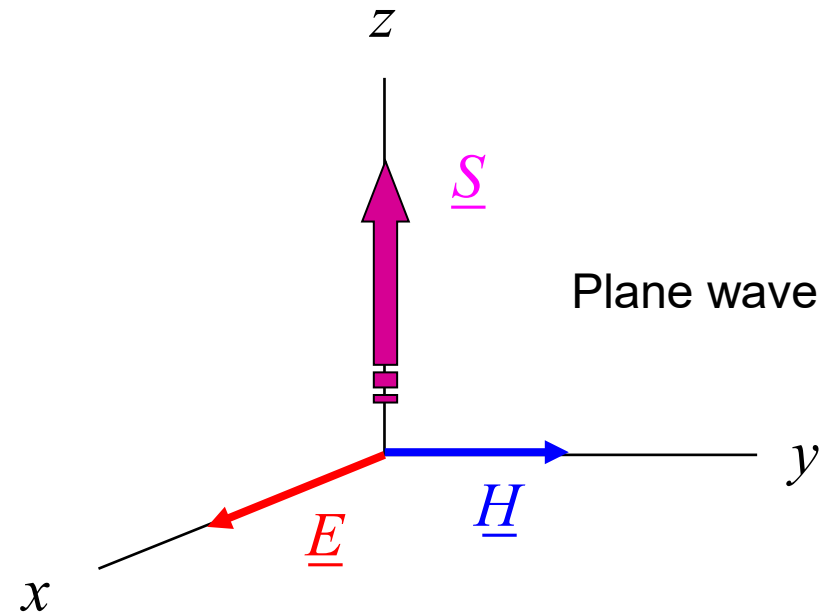
- A wave on a transmission line*
- A plane wave

* Exactly true if there is no conductor loss.

In each case the fields do not have a z component !



$$k_z = k$$



TEM_z Wave (cont.)

Relation between $\underline{E}$ and $\underline{H}$ for any TEM_z mode:

$$\underline{E} = -\eta (\hat{\underline{z}} \times \underline{H})$$

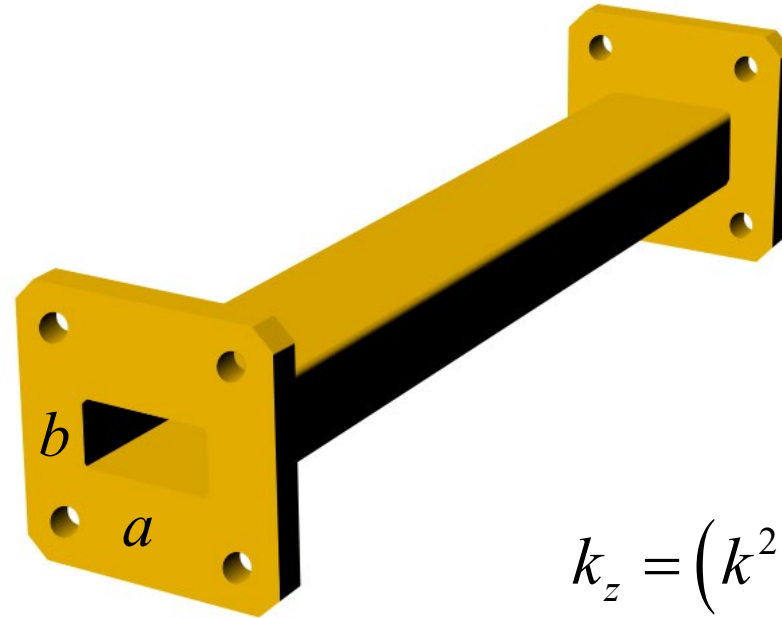
$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \eta_0 \sqrt{\frac{\mu_r}{\epsilon_r}} \quad (\eta_0 = 376.7303 \text{ } [\Omega]) \quad (\text{intrinsic impedance of filling material})$$

The electric and magnetic fields of a TEM_z wave are perpendicular to each other, and the amplitudes of them are related by η .

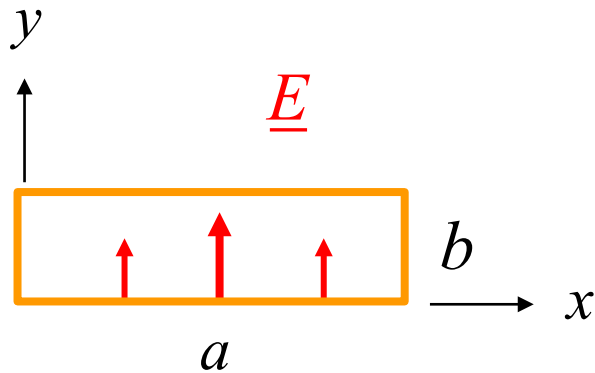
A derivation is given in Appendix C.

Rectangular Waveguide Modes

In a rectangular waveguide, the fields for any mode cannot be TEM_z.



$$k_z = \left(k^2 - k_c^2\right)^{1/2} \neq k$$



Rectangular Waveguide Modes (cont.)

In a rectangular waveguide (hollow pipe of metal), there are two types of modes:

$$\text{TM}_z: H_z = 0, \quad E_z \neq 0$$

$$\text{TE}_z: E_z = 0, \quad H_z \neq 0$$

Each type of mode can exist independently.

TM_z : Transverse magnetic field. The magnetic field is transverse to z (perpendicular to z).

TE_z : Transverse electric field. The electric field is transverse to z (perpendicular to z).

Rectangular Waveguide Modes (cont.)

Summary of TM_z Solution: TM_{mn} mode

$$E_z(x, y, z) = A_{mn} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-jk_z^{(m,n)} z}$$

$$k_z^{(m,n)} = \sqrt{k^2 - k_c^2} = \sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2} \quad k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

Note:

The other fields can be found from the table on slide 13.

$$m = 1, 2, \dots$$

$$n = 1, 2, \dots$$

Note: If either m or n is zero, the entire field is zero.

The TM mode with the lowest cutoff frequency is the TM_{11} mode.

$$f_c^{\text{TM}_{m,n}} = \frac{c_d}{2\pi} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \quad (\text{lossless waveguide})$$

Rectangular Waveguide Modes (cont.)

Summary of TE_z Solution: TE_{mn} mode

$$H_z(x, y, z) = A_{mn} \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-jk_z^{(m,n)}z}$$

$$k_z^{(m,n)} = \sqrt{k^2 - k_c^2} = \sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2} \quad k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

Note:

The other fields can be found from the table on slide 13.

$$m = 0, 1, 2, \dots$$

$$n = 0, 1, 2, \dots$$

$$(m, n) \neq (0, 0)$$

The TE mode with the lowest cutoff frequency is the TE_{10} mode (for $a > b$).

$$f_c^{\text{TE}_{m,n}} = \frac{c_d}{2\pi} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

(lossless waveguide)

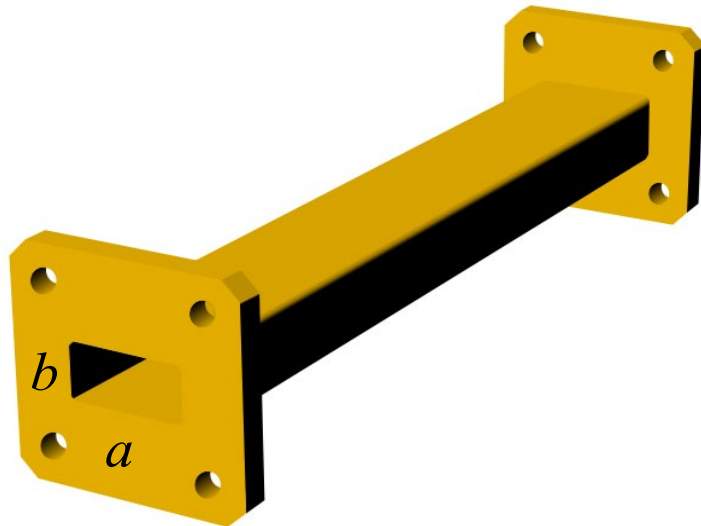
Note: Same formula for cutoff frequency as the TM_z case!

Rectangular Waveguide Modes (cont.)

The TE_{10} mode is the “dominant mode”
(the one with the lowest cutoff frequency).

Assumption: $a > b$

(The widest dimension is labelled as a .)



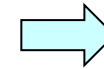
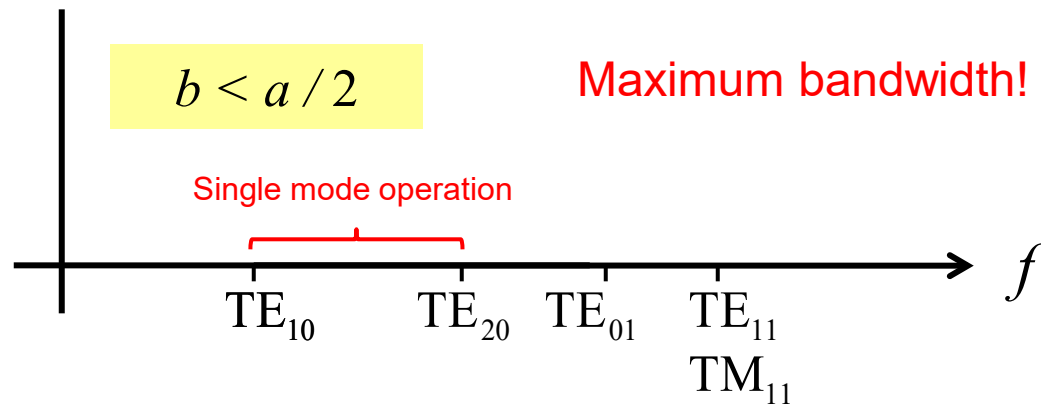
$$f_c^{\text{TE}_{10}} = \frac{c_d}{2a}$$

Recall:
$$f_c^{m,n} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

Rectangular Waveguide Modes (cont.)

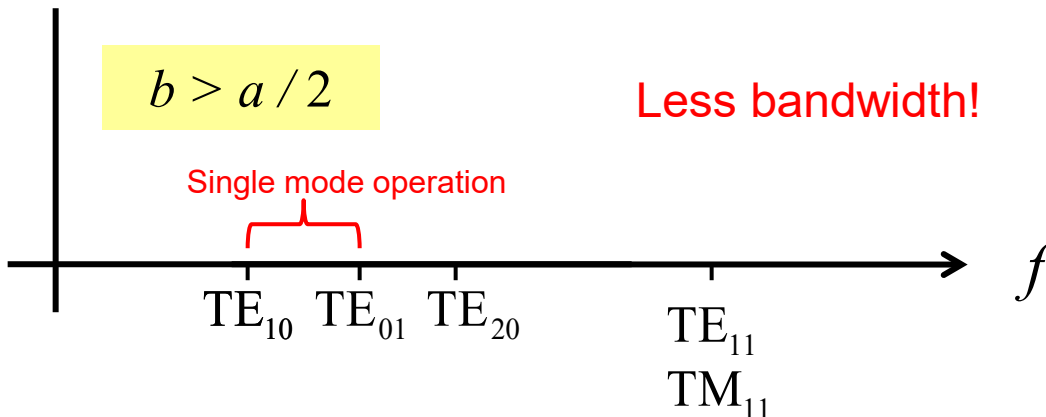
Two cases are considered:

Recall: $f_c^{m,n} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$



Operating range:

$$f_c^{TE_{10}} < f < 2f_c^{TE_{10}}$$



Dominant TE₁₀ Mode

Summary of TE₁₀ Mode

$$H_z(x, y, z) = A_{10} \cos\left(\frac{\pi x}{a}\right) e^{-jk_z z}$$

$$E_y(x, y, z) = E_{10} \sin\left(\frac{\pi x}{a}\right) e^{-jk_z z}$$

$$H_x(x, y, z) = -\left(\frac{k_z}{\omega\mu}\right) E_{10} \sin\left(\frac{\pi x}{a}\right) e^{-jk_z z}$$

where

$$E_{10} \equiv \left(-\frac{j\omega\mu}{k^2 - k_z^2}\right) \left(\frac{\pi}{a}\right) A_{10}$$

$$k_z = k_z^{(1,0)} = \sqrt{k^2 - \left(\frac{\pi}{a}\right)^2} \quad f_c^{\text{TE}_{1,0}} = \frac{c_d}{2a}$$

(lossless waveguide)

Dominant TE₁₀ Mode (cont.)

Power flow in lossless waveguide ($f > f_c$):

$$P_z = \left(\frac{ab}{4\omega\mu} \right) \beta |E_{10}|^2 \quad [\text{W}] \quad (\text{watts flowing down the waveguide})$$

$$\begin{aligned} P_z &= \text{Re} \int_0^b \int_0^a \frac{1}{2} (\underline{E} \times \underline{H}^*) \cdot \hat{z} \, dx dy \\ &= \text{Re} \int_0^b \int_0^a -\frac{1}{2} (E_y H_x^*) \, dx dy \\ &= -\frac{1}{2} b \text{Re} \int_0^a E_y H_x^* \, dx \end{aligned}$$

(The derivation is omitted, but please see the formula box above.)

Note: Above cutoff, there is only watts flowing (no vars). Below cutoff there is no watts flowing (only vars).

- ❖ Make b larger to get more power flow for a given value of a .
- ❖ Keep $b < a/2$ to avoid limiting the bandwidth due to the TE₀₁ mode (i.e., $f_c^{\text{TE}_{01}} > f_c^{\text{TE}_{20}}$)



The optimum dimension for b is $a/2$
(gives maximum power flow without sacrificing bandwidth).

Example

Find the single-mode operating frequency region for air-filled X-band waveguide.

Standard X-band* waveguide:

$$a = 0.900 \text{ inches (2.286 cm)}$$

$$b = 0.400 \text{ inches (1.016 cm)}$$

Use

$$f_c^{(1,0)} = \frac{c}{2a}$$

Hence, we have:

$$f_c^{(1,0)} = 6.56 \text{ [GHz]}$$

$$f_c^{(2,0)} = 13.11 \text{ [GHz]}$$

$$6.56 < f < 13.11 \text{ [GHz]}$$

Note: $b < a / 2$



X-band waveguide

* X-band: from 8.0 to 12 GHz.

Example (cont.)

- Find the phase constant of the TE₁₀ mode at 9.00 GHz.
- Find the attenuation in dB/m at 5.00 GHz

Recall: $f_c^{(1,0)} = 6.56 \text{ [GHz]}$



X-band waveguide

$$\beta = \sqrt{k^2 - \left(\frac{\pi}{a}\right)^2}, \quad f > f_c$$

$$\alpha = \sqrt{\left(\frac{\pi}{a}\right)^2 - k^2}, \quad f < f_c$$

$$k = k_0 = \omega \sqrt{\mu_0 \epsilon_0} = 2\pi f / c = 2\pi / \lambda_0$$

$$\text{At 9.0 GHz: } k = 188.62 \text{ [rad/m]}$$

$$\text{At 5.0 GHz: } k = 104.79 \text{ [rad/m]}$$

$$k_c = \pi / a = 137.43 \text{ [rad/m]}$$

$$\text{At 9.00 GHz: } \beta = 129.13 \text{ [rad/m]}$$

$$\text{At 5.00 GHz: } \alpha = 88.91 \text{ [nepers/m]}$$

Example (cont.)

At 5.0 GHz:

$$\alpha = 88.91 \text{ [nepers/m]}$$

Recall:

$$\text{dB/m} = 8.68589 \alpha$$

$$\text{Attenuation} = 772 \text{ dB/m}$$

This is a very rapid attenuation!



X-band waveguide

Waveguide Modes in Coax

Dominant waveguide mode in coax (derivation omitted):

TE₁₁ mode:
$$f_c^{\text{TE}_{11}} \approx \frac{c}{a\sqrt{\epsilon_r}} \left(\frac{1}{\pi} \right) \left(\frac{1}{1 + b/a} \right)$$

Note:
In this notation, the “11” subscript refers to the angular and radial variation.

Example: RG 142 coax

$$a = 0.035 \text{ inches} = 8.89 \times 10^{-4} \text{ [m]}$$

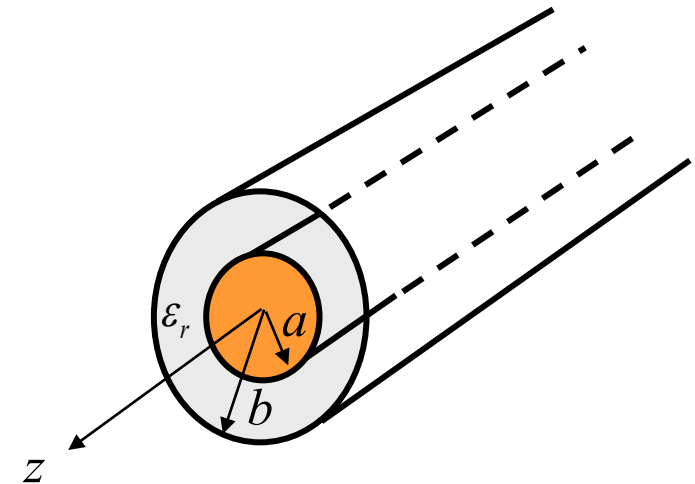
$$b = 0.116 \text{ inches} = 29.46 \times 10^{-4} \text{ [m]}$$

$$\epsilon_r = 2.2$$

$$b/a = 3.31 \Rightarrow Z_0 = 48.4 [\Omega]$$

$$f_c^{\text{TE}_{11}} \approx 16.8 \text{ [GHz]}$$

This coax cannot be used above 16.8 [GHz]



$$Z_0 = \sqrt{\frac{L}{C}} = \frac{\eta_0}{2\pi\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right)$$

Appendix A: Solution for Rectangular Waveguide Modes

TM_z Modes $H_z = 0, E_z \neq 0$

We want to solve:
$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \frac{\partial^2 E_z}{\partial z^2} + k^2 E_z = 0 \quad \Rightarrow \quad \frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + (k^2 - k_z^2) E_z = 0$$

Define: $k_c^2 \equiv k^2 - k_z^2$

We then have:
$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + k_c^2 E_z = 0$$

Note that k_c is an unknown at this point.

Dividing by the $\exp(-jk_z z)$ term, we have:

$$\frac{\partial^2 E_{z0}}{\partial x^2} + \frac{\partial^2 E_{z0}}{\partial y^2} + k_c^2 E_{z0} = 0$$

We solve the above equation by using the method of separation of variables.

We assume: $E_{z0}(x, y) = X(x) Y(y)$

Appendix A: Solution for Rectangular Waveguide Modes (cont.)

$$\frac{\partial^2 E_{z0}}{\partial x^2} + \frac{\partial^2 E_{z0}}{\partial y^2} + k_c^2 E_{z0} = 0$$

$$E_{z0}(x, y) = X(x) Y(y)$$

Hence $X''Y + XY'' = -k_c^2 XY$

Divide by XY : $\frac{X''}{X} + \frac{Y''}{Y} = -k_c^2$

Hence $\frac{X''}{X} = -k_c^2 - \frac{Y''}{Y}$

This has the following form: $F(x) = G(y)$

Both sides of the equation
must be a constant!

Appendix A: Solution for Rectangular Waveguide Modes (cont.)

$$\frac{X''}{X} = -k_c^2 - \frac{Y''}{Y} = \text{constant}$$

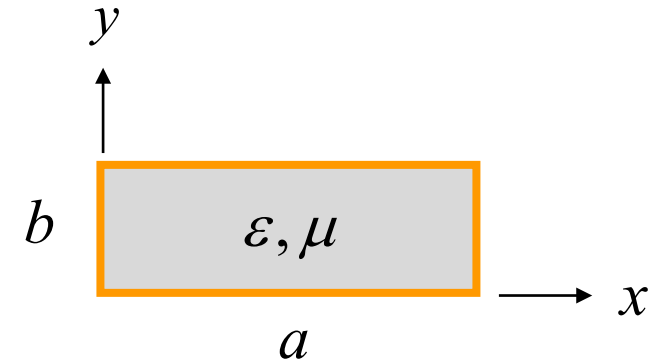
Denote $\frac{X''}{X} = \text{constant} \equiv -k_x^2$

General solution: $X(x) = A \sin(k_x x) + B \cos(k_x x)$

Boundary conditions: $\begin{cases} X(0) = 0 & (1) \\ X(a) = 0 & (2) \end{cases}$

$$(1) \Rightarrow B = 0 \Rightarrow X(x) = A \sin(k_x x)$$

$$(2) \Rightarrow \sin(k_x a) = 0$$



Appendix A: Solution for Rectangular Waveguide Modes (cont.)

From the last slide:

$$\sin(k_x a) = 0$$

This gives us the following result:

$$k_x a = m\pi, \quad m = 1, 2, \dots$$



$$k_x = \frac{m\pi}{a}$$

Hence, we have:

$$X(x) = A \sin\left(\frac{m\pi x}{a}\right)$$

Now we turn our attention to the $Y(y)$ function.

Appendix A: Solution for Rectangular Waveguide Modes (cont.)

We have:

$$\frac{X''}{X} = -k_c^2 - \frac{Y''}{Y} = -k_x^2$$

Hence: $\frac{Y''}{Y} = k_x^2 - k_c^2$

Denote: $k_y^2 \equiv k_c^2 - k_x^2$

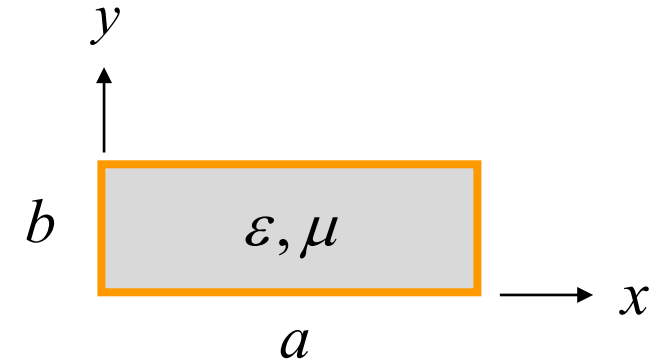
Then we have: $\frac{Y''}{Y} = -k_y^2$

General solution: $Y(y) = C \sin(k_y y) + D \cos(k_y y)$

Appendix A: Solution for Rectangular Waveguide Modes (cont.)

$$Y(y) = C \sin(k_y y) + D \cos(k_y y)$$

Boundary conditions:
$$\begin{cases} Y(0) = 0 & (3) \\ Y(b) = 0 & (4) \end{cases}$$



$$(3) \Rightarrow D = 0 \Rightarrow Y(y) = C \sin(k_y y)$$

$$(4) \Rightarrow \sin(k_y b) = 0$$

Equation (4) gives us the following result: $k_y b = n\pi, \quad n = 1, 2, \dots$

$$\Rightarrow k_y = \frac{n\pi}{b}$$

Appendix A: Solution for Rectangular Waveguide Modes (cont.)

The $Y(y)$ function is then: $Y(y) = C \sin\left(\frac{n\pi y}{b}\right)$

Therefore, we have:

$$E_{z0}(x, y) = X(x)Y(y) = AC \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)$$

New notation:

$$E_{z0}(x, y) = A_{mn} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)$$

The E_z field inside the waveguide thus has the following form:

$$E_z(x, y, z) = A_{mn} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-jk_z^{(m,n)}z}$$

Appendix A: Solution for Rectangular Waveguide Modes (cont.)

Recall: $k_y^2 = k_c^2 - k_x^2$

Hence, we have: $k_c^2 = k_x^2 + k_y^2$

Therefore, the solution for k_c is given by:

$$k_c^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

Next, recall that $k_c^2 = k^2 - k_z^2$

Hence

$$k_z^2 = k^2 - k_c^2 \quad \Rightarrow$$

$$k_z = k_z^{(m,n)} = \sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

Appendix A: Solution for Rectangular Waveguide Modes (cont.)

TE_z Modes

$$E_z = 0, \quad H_z \neq 0$$

We now start with:

$$\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + (k^2 - k_z^2) H_z = 0$$

Using the separation of variables method again, we have:

$$H_{z0}(x, y) = X(x) Y(y)$$

where

$$X(x) = A \sin(k_x x) + B \cos(k_x x)$$

$$Y(y) = C \sin(k_y y) + D \cos(k_y y)$$

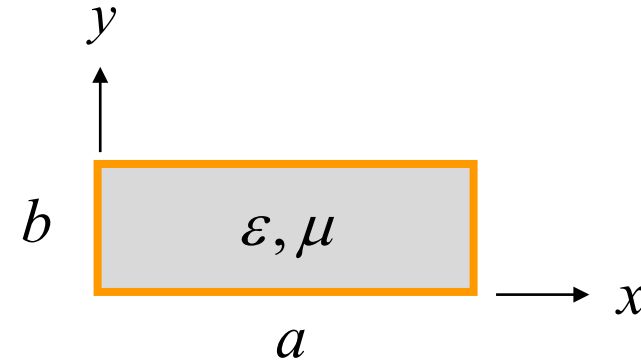
and

$$k_c^2 = k_x^2 + k_y^2 \quad k_z^2 = k^2 - k_c^2$$

Appendix A: Solution for Rectangular Waveguide Modes (cont.)

Boundary conditions:

$$\begin{aligned} E_x(x, 0) &= 0 & E_y(0, y) &= 0 \\ E_x(x, b) &= 0 & E_y(a, y) &= 0 \end{aligned}$$



The result is:

$$H_{z0}(x, y) = A_{mn} \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)$$

This can be shown by using the following equations:

$$\begin{aligned} E_x &= \left(\frac{-j\omega\mu}{k^2 - k_z^2}\right) \frac{\partial H_z}{\partial y} - \left(\frac{jk_z}{k^2 - k_z^2}\right) \cancel{\frac{\partial E_z}{\partial x}} \quad \Rightarrow \quad \frac{\partial H_z}{\partial y} = 0, \quad y = 0, b \\ E_y &= \left(\frac{j\omega\mu}{k^2 - k_z^2}\right) \frac{\partial H_z}{\partial x} - \left(\frac{jk_z}{k^2 - k_z^2}\right) \cancel{\frac{\partial E_z}{\partial y}} \quad \Rightarrow \quad \frac{\partial H_z}{\partial x} = 0, \quad x = 0, a \end{aligned}$$

Appendix A: Solution for Rectangular Waveguide Modes (cont.)

The H_z field inside the waveguide thus has the following form:

$$H_z(x, y, z) = A_{mn} \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-jk_z^{(m,n)}z}$$

$$k_z^{(m,n)} = \sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

Same formula for cutoff frequency as the TE_z case!

$$\begin{aligned} m &= 0, 1, 2, \dots \\ n &= 0, 1, 2, \dots \end{aligned} \quad (m, n) \neq (0, 0)$$

Note: The (0,0) TE_z mode is not valid, since it violates the magnetic Gauss law:

$$\underline{H}(x, y, z) = \hat{z} A_{00} e^{-jkz} \quad \Rightarrow \quad \nabla \cdot \underline{H}(x, y, z) \neq 0$$

Appendix B

Derivation of Field Expression Table

(illustrated for E_y)

$$\nabla \times \underline{H} = j \omega \epsilon \underline{E}$$

$$\Rightarrow E_y = \frac{1}{j\omega\epsilon} \left(-\frac{\partial H_z}{\partial x} + \frac{\partial H_x}{\partial z} \right)$$

or

$$E_y = \frac{1}{j\omega\epsilon} \left(-\frac{\partial H_z}{\partial x} - jk_z H_x \right)$$

Now solve for H_x :

$$\nabla \times \underline{E} = -j \omega \mu \underline{H}$$

Appendix B (cont.)

$$\nabla \times \underline{E} = -j \omega \mu \underline{H}$$

$$\begin{aligned} \Rightarrow H_x &= -\frac{1}{j\omega\mu} \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \\ &= -\frac{1}{j\omega\mu} \left(\frac{\partial E_z}{\partial y} + jk_z E_y \right) \end{aligned}$$

Substituting this into the equation for E_y yields:

$$E_y = \frac{1}{j\omega\epsilon} \left[-\frac{\partial H_z}{\partial x} - jk_z \left[\left(-\frac{1}{j\omega\mu} \right) \left(\frac{\partial E_z}{\partial y} + jk_z E_y \right) \right] \right]$$

Next, multiply by $-j\omega\mu(j\omega\epsilon) = k^2$

Appendix B (cont.)

This gives us:

$$k^2 E_y = j\omega\mu \frac{\partial H_z}{\partial x} - jk_z \frac{\partial E_z}{\partial y} + k_z^2 E_y$$

Combining terms, we have:

$$(k^2 - k_z^2) E_y = j\omega\mu \frac{\partial H_z}{\partial x} - jk_z \frac{\partial E_z}{\partial y}$$

Solving for E_y , we have:

$$E_y = \left(\frac{j\omega\mu}{k^2 - k_z^2} \right) \frac{\partial H_z}{\partial x} - \left(\frac{jk_z}{k^2 - k_z^2} \right) \frac{\partial E_z}{\partial y}$$

The other three components E_x , H_x , H_y may be found similarly.

Appendix C

Relation Between E and H for any TEM_z Mode

Faraday's Law: $\nabla \times \underline{E} = -j\omega\mu\underline{H}$

Take the x component of both sides: $\frac{\partial \cancel{E_z}}{\partial y} - \frac{\partial E_y}{\partial z} = -j\omega\mu H_x$

Assume that the field varies as $E_y(x, y, z) = E_{y0}(x, y)e^{-jkz}$

Hence, $-(-jk)E_y = -j\omega\mu H_x$

Therefore, we have: $\frac{E_y}{H_x} = -\frac{\omega\mu}{k} = -\frac{\omega\mu}{\omega\sqrt{\mu\varepsilon}} = -\sqrt{\frac{\mu}{\varepsilon}} = -\eta$

Appendix C (cont.)

$$\nabla \times \underline{E} = -j\omega\mu\underline{H}$$

Now take the y component of both sides: $-\frac{\cancel{\partial E_z}}{\partial x} + \frac{\partial E_x}{\partial z} = -j\omega\mu H_y$

Hence, we have: $(-jk)E_x = -j\omega\mu H_y$

Therefore, we have $\frac{E_x}{H_y} = \frac{\omega\mu}{k} = \frac{\omega\mu}{\omega\sqrt{\mu\epsilon}} = \sqrt{\frac{\mu}{\epsilon}} = \eta$

Therefore, we have: $\frac{E_x}{H_y} = \eta$

Appendix C (cont.)

Summary: $\frac{E_y}{H_x} = -\eta \quad \frac{E_x}{H_y} = \eta$

These two equations may be written as a single vector equation:

$$\underline{E} = -\eta (\hat{z} \times \underline{H})$$

The electric and magnetic fields of a TEM_z wave are perpendicular to each other, and the amplitudes of them are related by η .